

Urban Noise Mapping — Hanoi

Data Analysis Report

220 smartphone measurements · 3 sites · 10 Jun – 30 Jun 2026 2026

Methodology adapted from Sunbird AI (Urban Noise Uganda 61K)

220

Measurements

3

Sites

47-88

dB range

38%

Exceed QCVN

17:00

Peak hour

1. Data collection summary

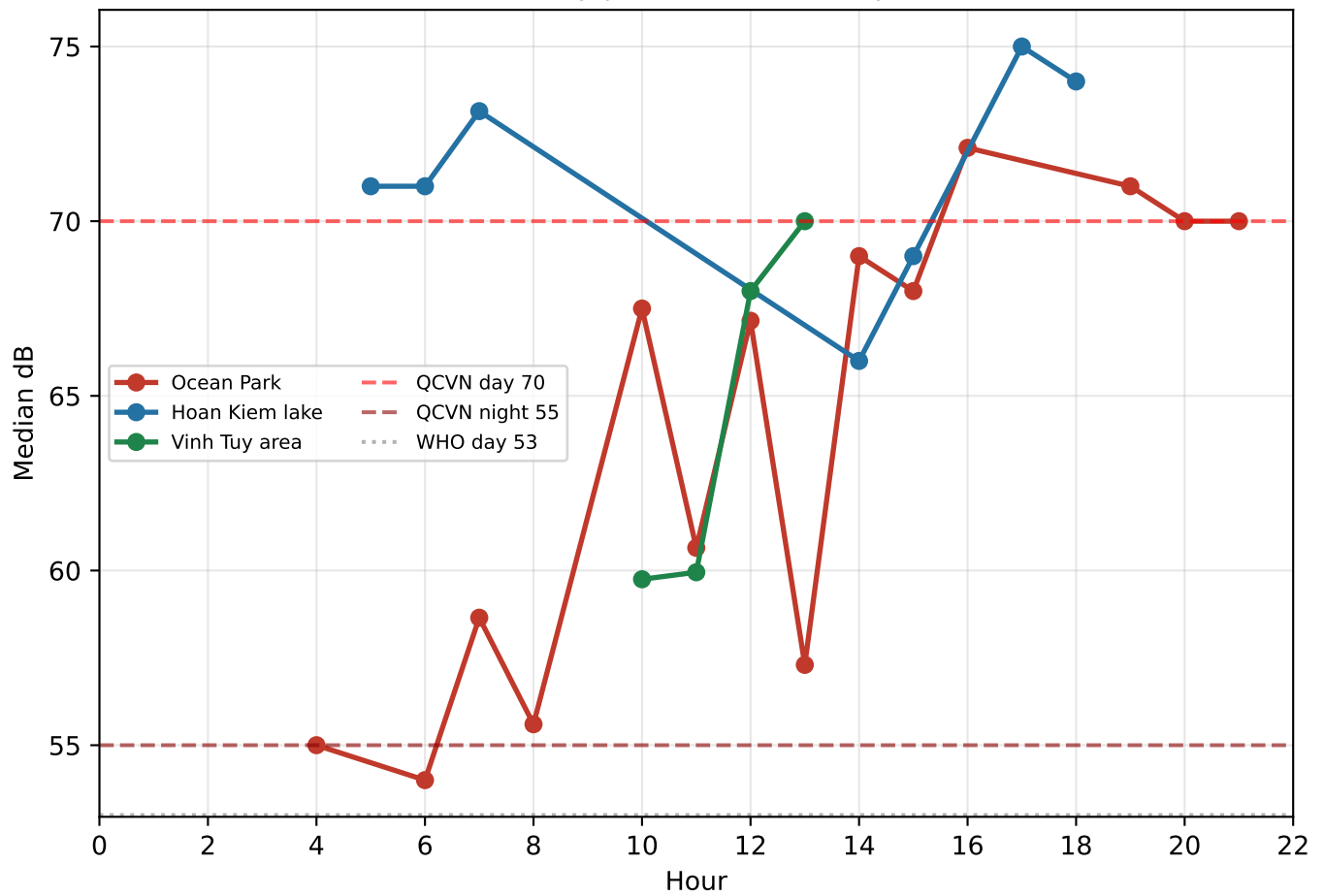
Site	n	Median dB	Min-Max	% < 60 dB	% roadside
Ocean Park	148	65	47-88	36%	68%
Hoan Kiem lake	44	72	57-80	2%	82%
Vinh Tuy area	28	62	53-73	36%	50%
ALL	220	66	47-88	29%	68%

2. Protocol & standards

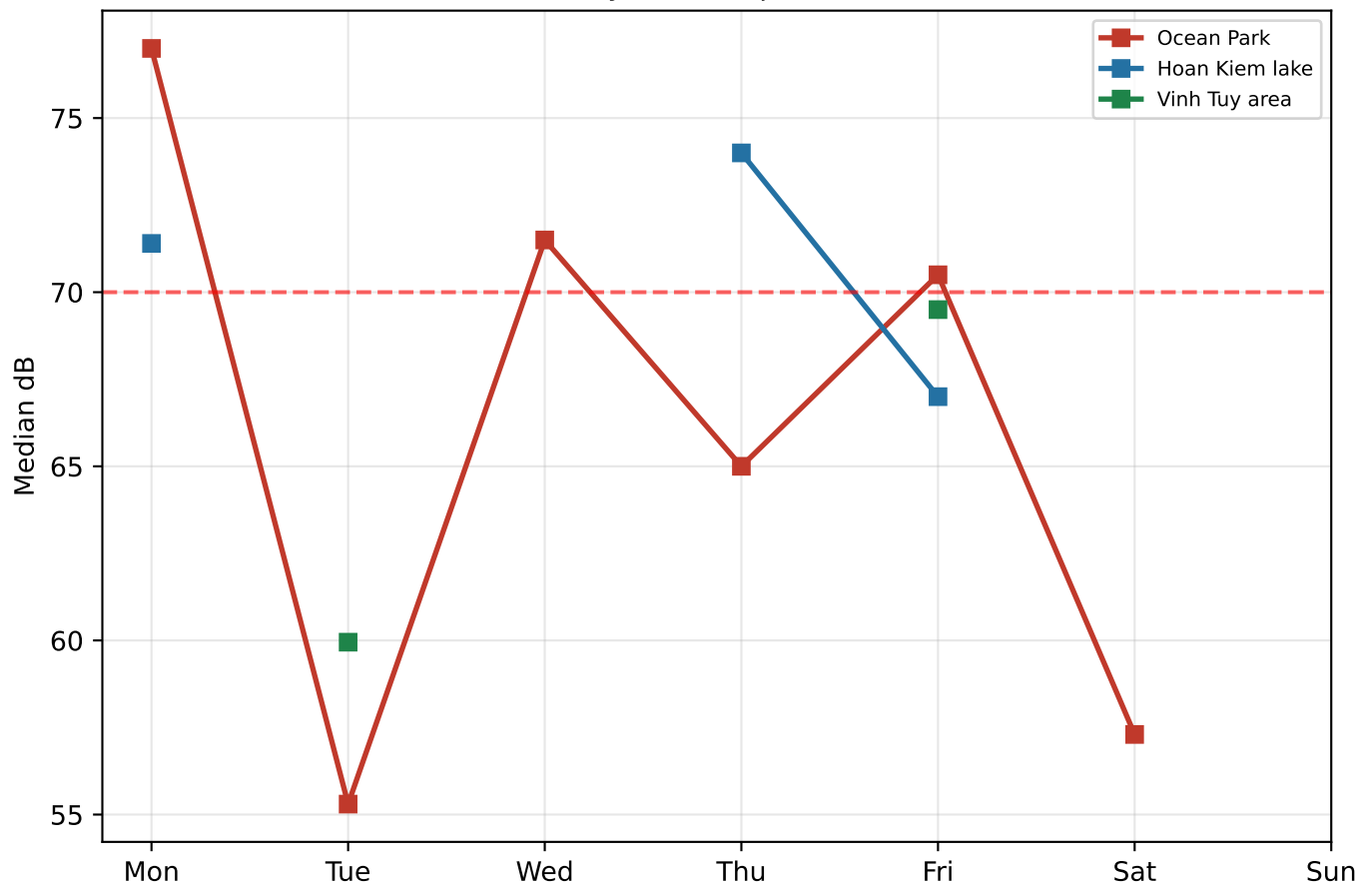
- Capture: smartphone (Decibel X, A-weighting), ODK Collect + KoboToolbox, walking sampling, GPS < 10 m accuracy, >=10 s audio per point, cross-calibrated phones, 05:00-23:00 window.
- Per point: location, dB, dominant source category, distance to road, audio; construction sites logged separately (radius transect). 40 traffic videos for later vehicle counting.
- Standards: QCVN 26:2010/BTNMT = 70 dB day (6-21h) / 55 dB night. WHO = 53 / 45 dB.

3. Temporal patterns by site

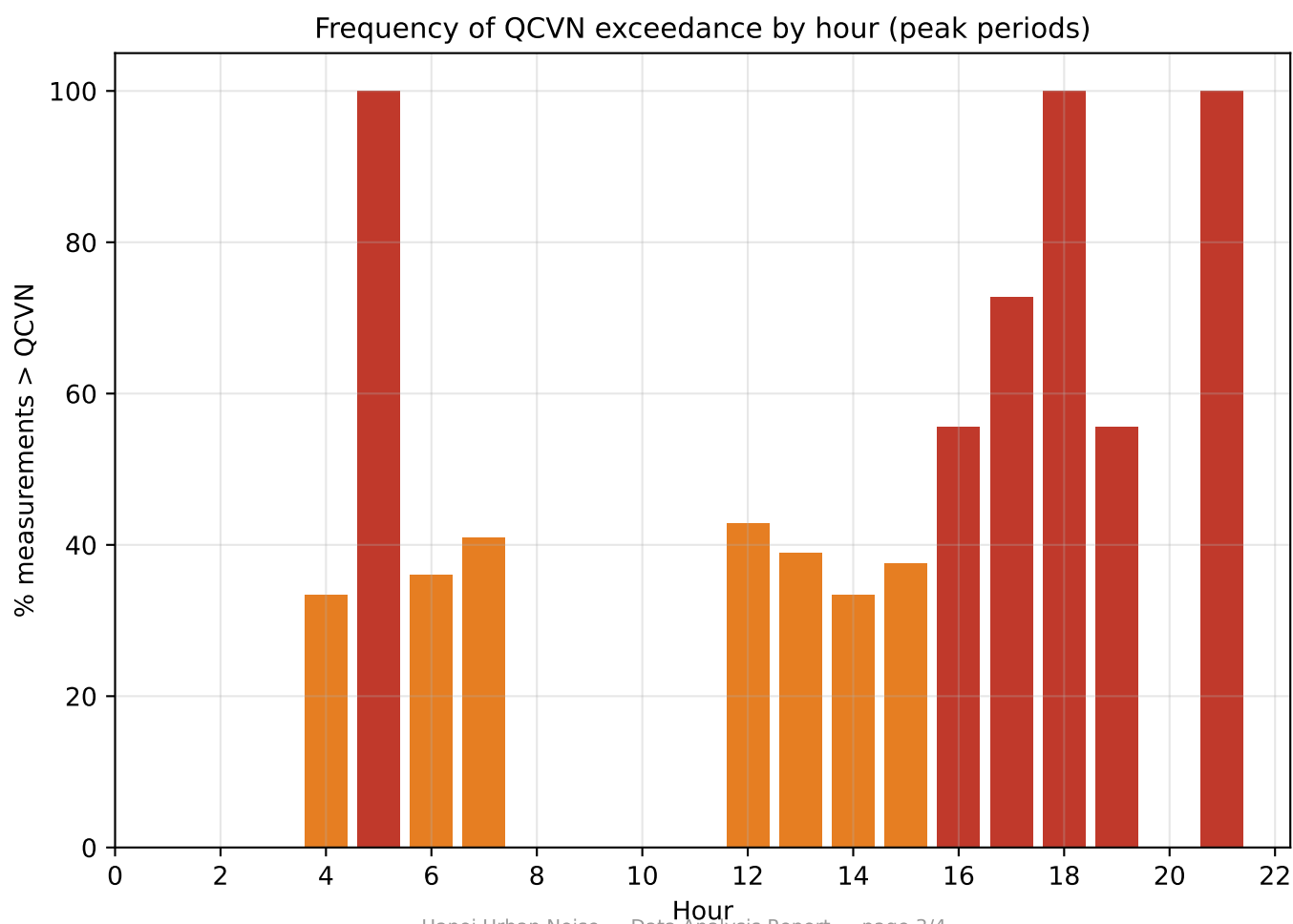
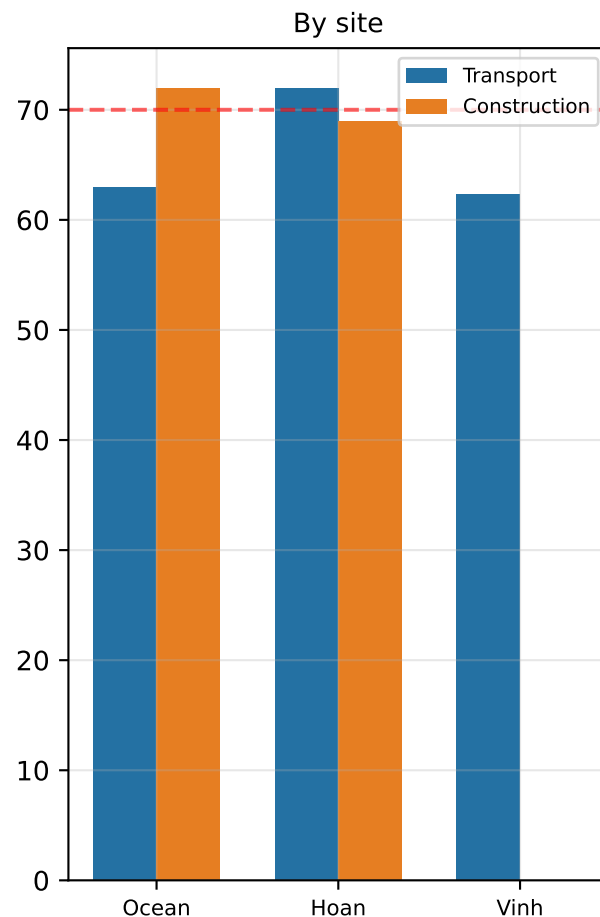
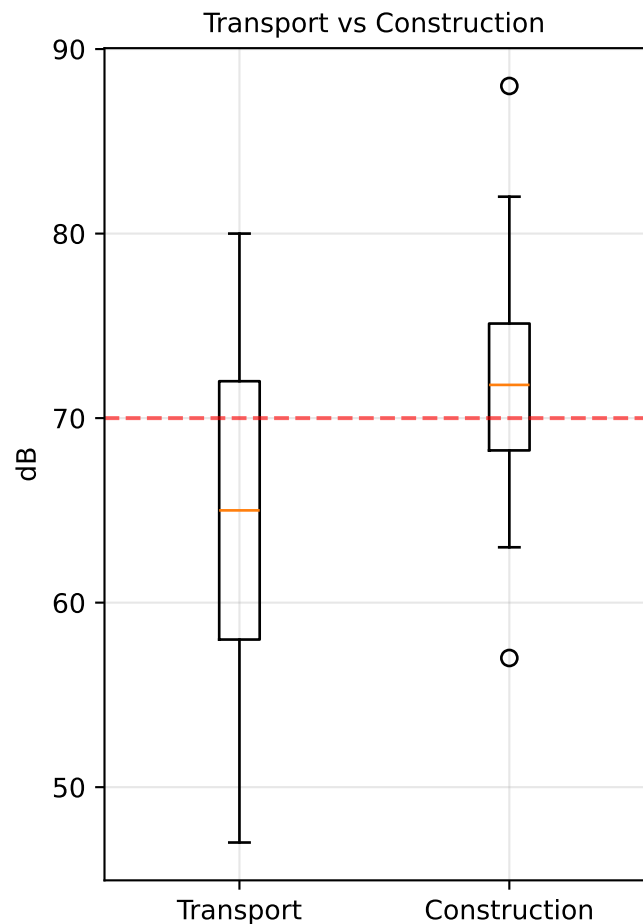
Hourly pattern within a day



Day-of-week pattern



4. Noise sources & standard exceedances



5. Predictive model (morphology + time -> dB)

LightGBM on urban-morphology features (built ratio, road density, distance to road, intersections) + hour + weekend. Reported on held-out data.

	Method	r	R ²	MAE (dB)
Direct training (spatial CV, hon		0.69	0.45	4.7
Direct training (random split, upper		0.74	0.53	4.3
Uganda->Hanoi transfer (compa		0.23	-0.68	8.5
Barcelona reference (profs)		0.66	0.61	—

Per-site generalization (leave-one-site-out): Ocean Park R² 0.46 · Vinh Tuy R² 0.18 · Hoan Kiem R² -0.89

6. QCVN 26:2010 exceedances by site

Site	Period	n	Median dB	Limit	% exceed	Mean excess
Ocean Park	day	144	65	70	33%	5.0 dB
Ocean Park	night	4	57	55	50%	9.5 dB
Hoan Kiem lake	day	38	72	70	68%	4.6 dB
Hoan Kiem lake	night	6	71	55	100%	15.2 dB
Vinh Tuy area	day	28	62	70	7%	2.0 dB

7. Limitations

- The model generalizes well within consistently-sampled sites (Ocean Park R² 0.46) but poorly to the historic centre (Hoan Kiem), where sampling was narrow (mostly roadside, little dB variation) — a sampling issue, not a model limitation.
- Instantaneous smartphone readings carry ~±5 dB of irreducible noise (a passing bus), which caps the achievable R² (~0.6); aggregating repeated measurements would help.
- Cross-city transfer (Uganda -> Hanoi) failed (R² < 0); only direct local training works.
- Non-professional sound meter: a constant calibration bias is correctable, clipping > 90 dB and wind are not. Weekend loud points and a third quiet site remain under-sampled.

Next: collect varied quieter points at Hoan Kiem; vehicle counting via computer vision on traffic videos; GAMA simulation (calibrated noise map + traffic scenarios).